

under low light intensities. Organic photovoltaic materials, valued for their flexibility and lightweight nature, can be tuned to absorb specific light wavelengths, thereby enhancing efficiency with indoor lighting. Dye-sensitised solar cells (DSSCs) excel in low-light conditions by using organic dyes to capture light, optimising efficiency for indoor lighting spectra. Recent advancements in perovskite materials have shown high absorption efficiency and tunability to indoor light wavelengths, making them a promising option for indoor solar applications.

Size and integration. Indoor solar cells are becoming smaller and more adaptable, facilitating seamless integration into everyday objects. This includes everything from smart home devices and sensors to wearable technology.

How are they different from earlier cells?

The primary difference between indoor solar cells and their predecessors lies in their design and functionality:

Light sensitivity. Earlier solar cells were optimised for outdoor sunlight, making them less effective under artificial lighting. Indoor solar cells, on the other hand, are specifically tuned to work under low-light conditions, enhancing their efficiency indoors.

Material efficiency. The use of advanced materials like perovskites has significantly improved the efficiency and durability of indoor solar cells compared to older silicon-based cells.

Application versatility. While traditional solar cells were mostly confined to rooftops and solar farms, indoor solar cells can be integrated into various small-scale applications, from powering IoT devices to supporting smart home systems.

Innovators in the field

As the demand for renewable energy grows, companies and research institutions are advancing solar technology. Heliatek in Germany leads in organic

Factors influencing the performance of indoor solar cells

1. Light intensity. Indoor light intensity is much lower than sunlight, typically around 100-500 lux compared to 100,000 lux for direct sunlight. Solar cells must be highly sensitive to low light levels to perform well indoors.

2. Spectral response. The spectral distribution of indoor light sources differs significantly from sunlight. Indoor solar cells need to be optimised for the specific wavelengths emitted by artificial lighting, such as LEDs or fluorescents, which have distinct spectral peaks.

3. Material properties. The choice of materials affects the cell's ability to absorb and convert light efficiently. Materials with broad absorption spectra and high quantum efficiency at low light intensities are preferred for indoor applications.

solar cells, effective in low light and versatile for surfaces like windows and walls. The UK's GCell enhances dye-sensitised solar cells (DSSCs) for indoor use. MIT's Research Laboratory of Electronics is improving indoor solar cell efficiency with advanced materials. Panasonic focuses on perovskite solar cells for indoor and outdoor applications, aiming for versatile, efficient energy solutions.

Recent innovations include organic photovoltaic cells that efficiently convert low-intensity light and perovskite solar cells known for high efficiency and low-cost production. New products integrate solar cells into smart sensors, wearable devices, and wireless chargers. Flexible and transparent solar cells are also being developed for seamless integration into various surfaces, enhancing functionality and aesthetics.

Leading companies like Panasonic and Sharp, alongside startups like Ambient Photonics, SolarWindow Technologies, Exeger, and Heliatek, are advancing indoor solar cell technology with innovative applications and high-efficiency products, driving market growth. Bifacial solar cells (BFSCs) generate more power by absorbing sunlight from both sides. Third-generation PV solar cells, known for flexibility and colour tunability, can also be designed as BFSCs.

In April 2024, a new series of hydrogenated amorphous silicon (a-Si:H) solar cells was unveiled, offering a 20% efficiency increase. These cells feature a black-reflective surface, thinner electrodes, and increased

layer thickness, enhancing efficiency. They are ideal for energy-harvesting applications like IoT devices, watches, sensor nodes, asset trackers, and remote controls, performing well in low and artificial light. This development aligns with sustainability and energy efficiency trends, meeting various industry needs with their high efficiency and compact design.

Transformative impact of indoor solar cell innovations

Innovations in indoor solar cell technology are driving a wave of new applications, revolutionising the way energy usage is perceived indoors:

Smart homes. Indoor solar cells can power various smart home devices, such as sensors, security systems, automated lighting, and climate control, reducing reliance on traditional power sources. For example, a smart home in California uses organic photovoltaic cells in window blinds to capture ambient light, reducing grid electricity use and ensuring continuous operation of critical devices during power outages.

Wearable technology. Integrating solar cells into wearables like fitness trackers and smartwatches can extend battery life and enhance user convenience.

Industrial IoT. Indoor solar cells in industrial settings provide a reliable power source for IoT devices, improving efficiency and reducing maintenance costs. A tech company in Tokyo has embedded solar cells in light fixtures and furniture, powering low-energy devices like wireless keyboards, mice, and desk lamps, result-